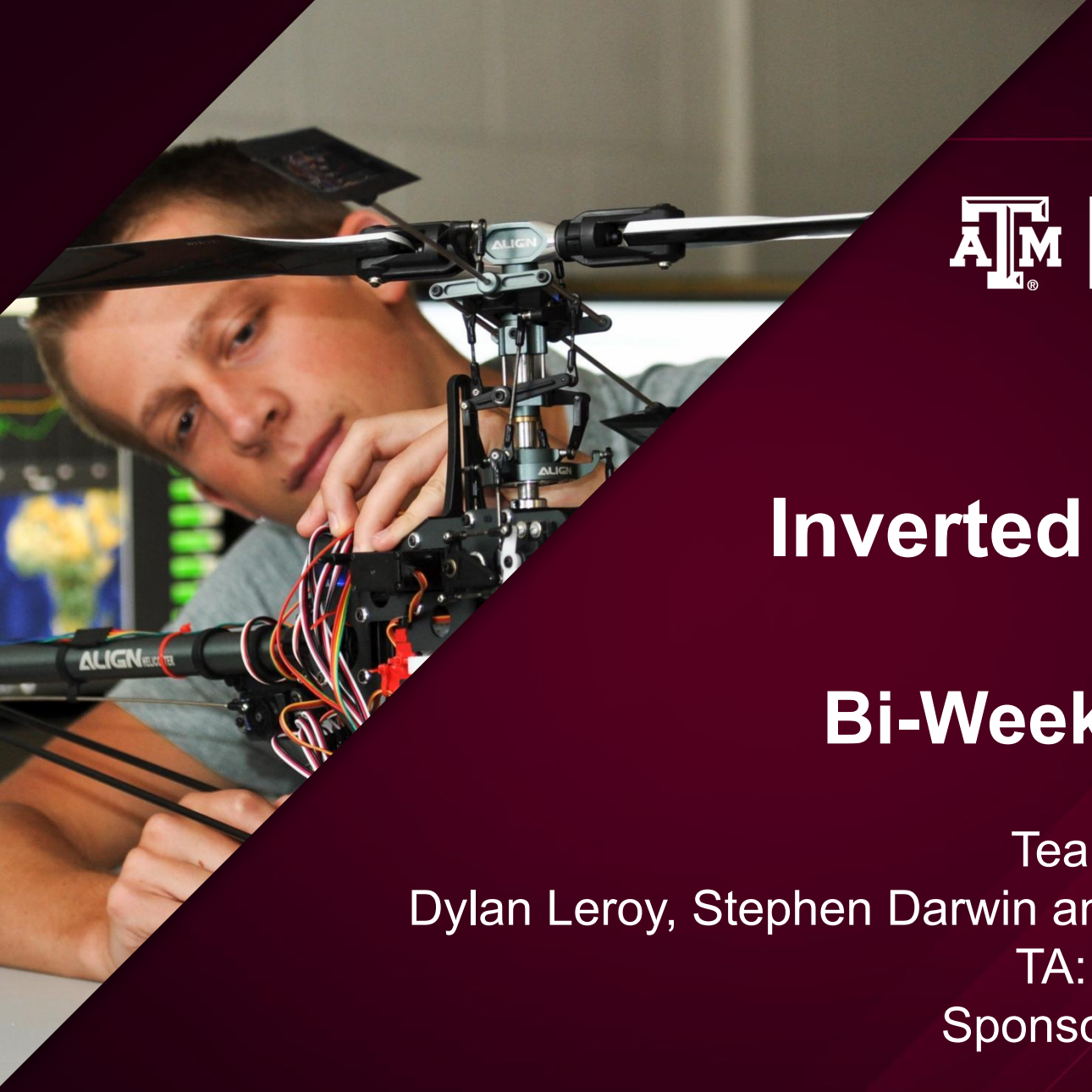

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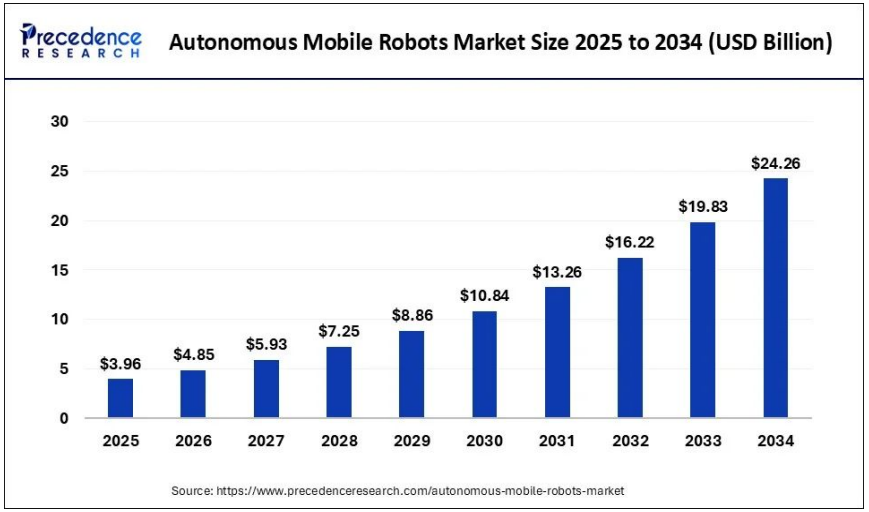
ECEN 404
Inverted Pendulum Robot 3
Bi-Weekly Update 2

Team Number: 55
Team Lead: Sean Wall
Dylan Leroy, Stephen Darwin and Arian Golkarfard
TA: Sabyasachi Gupta
Sponsor: Aniruddha Datta

Project Summary

Problem statement:

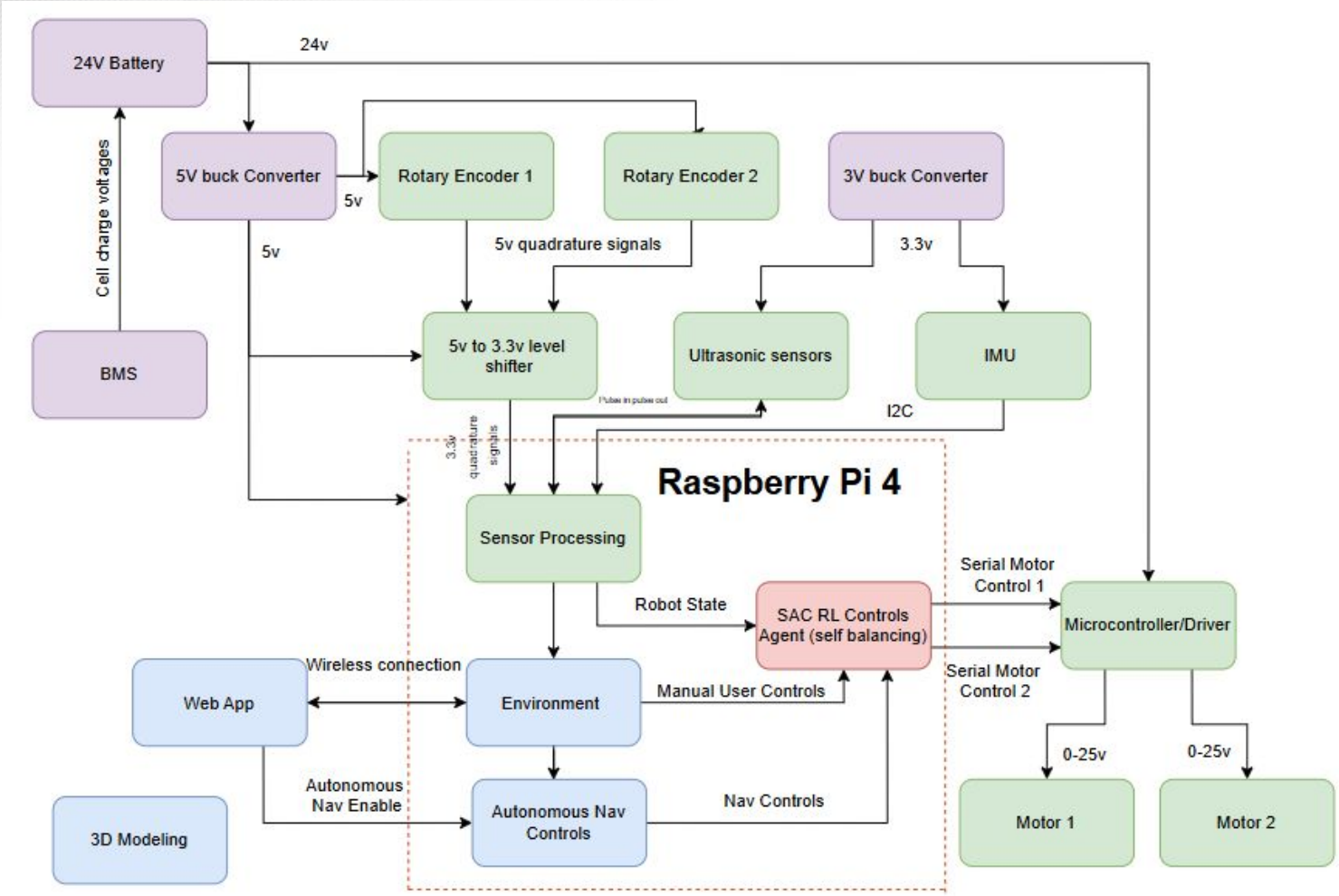
- Current mobile robot applications are rigid and inflexible
- Warehouse environments could benefit from more robust transportation methods for lightweight packages



Our Solution:

- Use reinforcement learning algorithm to train two-wheeled balancing robot
- RL will allow for a more robust and adaptable control system

System Overview



Project Timeline

Finish Wiring the robot (2/7)	Integration o the Sensor and App Subsystem (2/9)	Integration of the RL subsystem with the Database (2/11)	Boards arrived and soldered (to complete by 3/2)	Full Test (to complete by 3/5)	Movement/ Training Completed (to complete by 3/9)	Demo and Report (to complete by 4/14)

Sensor & Motor Controller

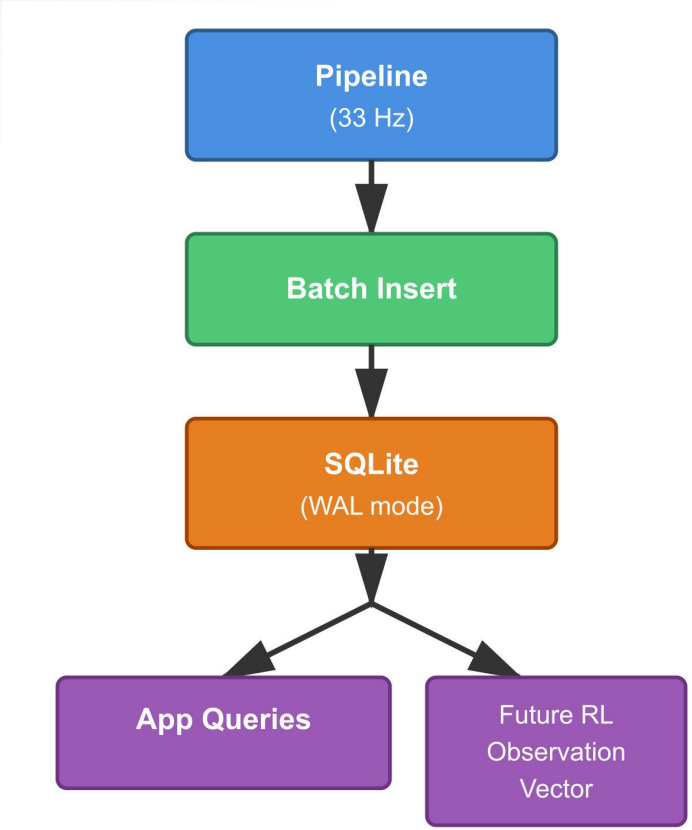
Arian Golkarfard

Accomplishments since last update 10 hrs of effort	Ongoing progress/problems and plans until the next presentation
• Completed SQLite migration: pipeline now writes IMU, encoder, and kinematics data to a database for app and RL integration • Added query utilities so the app can read from the database; pipeline runs at ~33 Hz • Validated and tested 2 front ultrasonic sensors and ready to integrate to the main pipeline	• Integrating 2 front ultrasonics into the single pipeline (IMUs + encoders) with no interference • Address and fix motor controller voltage leak • Integrating the Motor controller output with sensor input

Sensor Feedback

Arian Golkarfard

Main sensor pipeline data flow:



Data from SQLite ready for app read and RL observation vector:

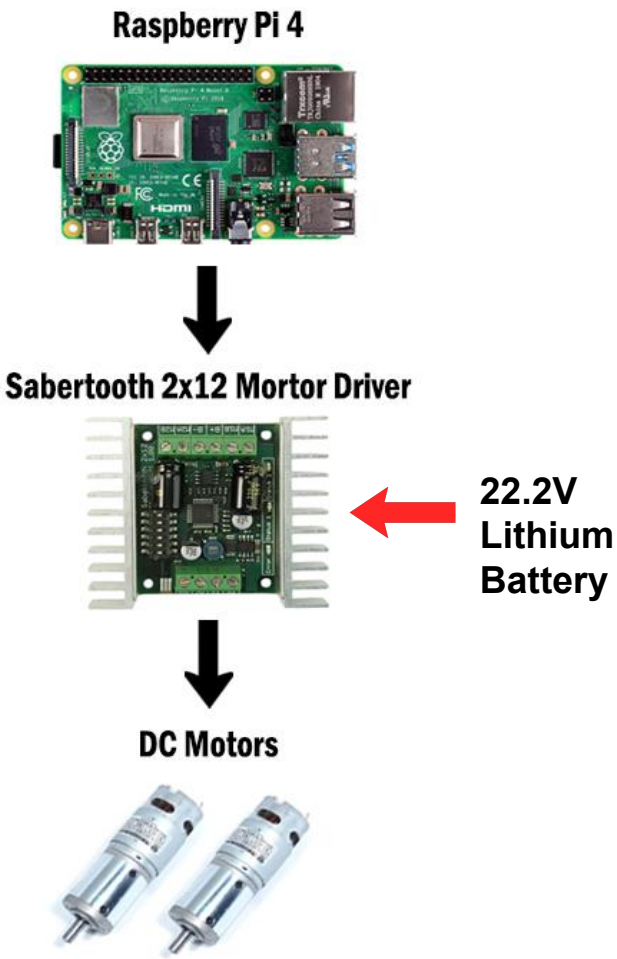
```
pendulum@Pendulum:~/Downloads $ python3 read_last_readings.py 1 --full
Last 1 readings (most recent first) - FULL VIEW:

================================================================================

--- Reading #1 ---
Timestamp: 2026-02-04 18:23:25.511798 (2800.054)

  IMU1 (Chassis):
    Gyro:        gx=0.0000, gy=0.0000, gz=0.0000 rad/s
    Accel:       ax=0.0586, ay=-0.1310, az=0.0090 m/s²
    Body Pitch:  1.4214 rad (81.44°)
    Yaw Rate:    0.0000 rad/s
    Rot Vel:     0.0000 rad/s
    Velocity:    vx=0.0000, vy=0.0000, vz=0.0000 m/s

  IMU2 (Pendulum):
    Gyro:        gx=0.0000, gy=0.0000, gz=0.0000 rad/s
    Accel:       ax=0.0000, ay=0.0000, az=0.0000 m/s²
    Pend Angle:  0.0000 rad (0.00°)
    Pend AngVel: 0.0000 rad/s
    Rot Vel:     0.0000 rad/s

  Encoders:
    Left:  0.0000 rad/s  (stopped)
    Right: 0.0000 rad/s  (stopped)

  Robot Kinematics:
    Forward vel: 0.0000 m/s
    Yaw rate:    0.0000 rad/s

  Status:
    IMU1: Connected
```

Motor Controller

Arian Golkarfard

Sabertooth 2x12: Dual-channel motor control for up two DC motors at 12 A

Current focus: Addressing motor controller voltage leak; once fixed, same PWM interface will be driven by SAC agent.



RL Controls System

Dylan Leroy

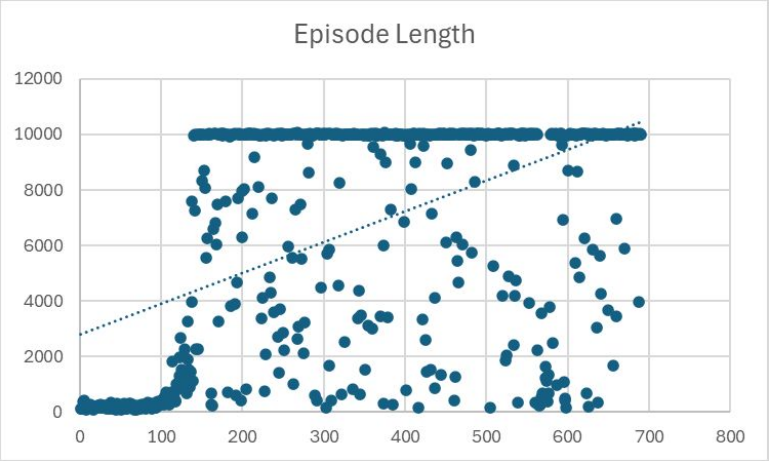
Accomplishments since last update 10 hrs of effort	Ongoing progress/problems and plans until the next presentation
• Integrated vector normalization into training which solved long term training issue • Validated proper training with ~2 million training steps	• Finalize training • Start to implement training on hardware

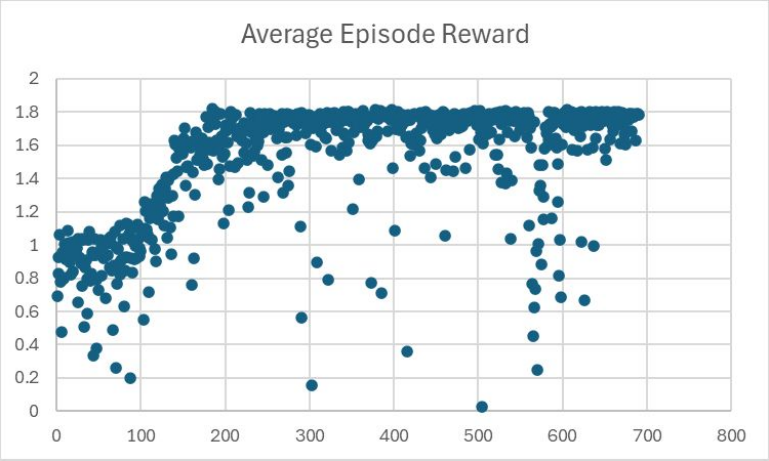
RL Controls

Dylan LeRoy

Subsystem Progress:

- Integrated vector normalization into training logic
  - Keeps a running mean and variance of all observations for normalization as training progresses
- Validated that changes fixed persistent training issue
  - Trained for ~2 million steps, degradation usually happened at around 5-600k steps

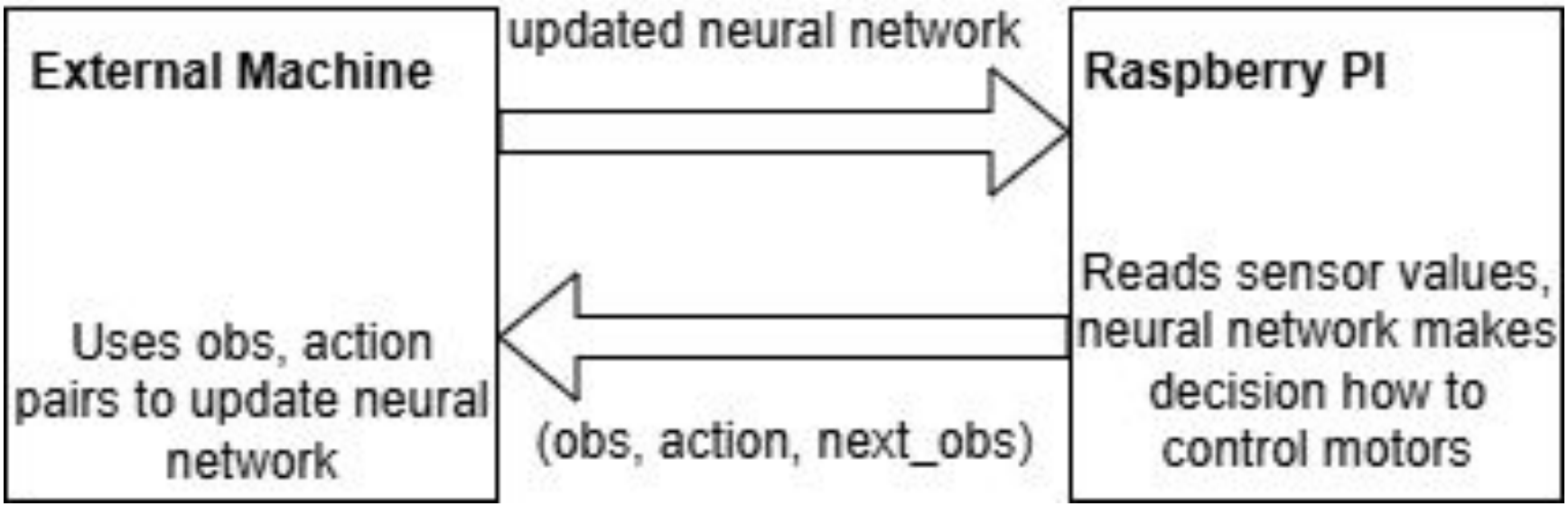




RL Controls

Dylan LeRoy

Subsystem Integration:



Web App

Sean Wall

Accomplishments since last update 10 hrs of effort	Ongoing progress/problems and plans until the next presentation
• Ordered the new PCBs and Components • Wired most of the robot to allow for integration	• Integrating the app to the SQLite database • Ensuring that the Raspberry Pi can receive data with the new wiring

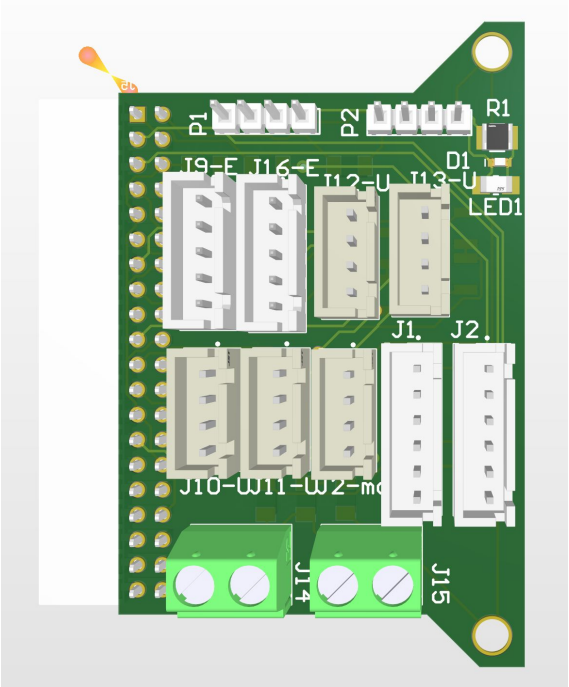
Web App

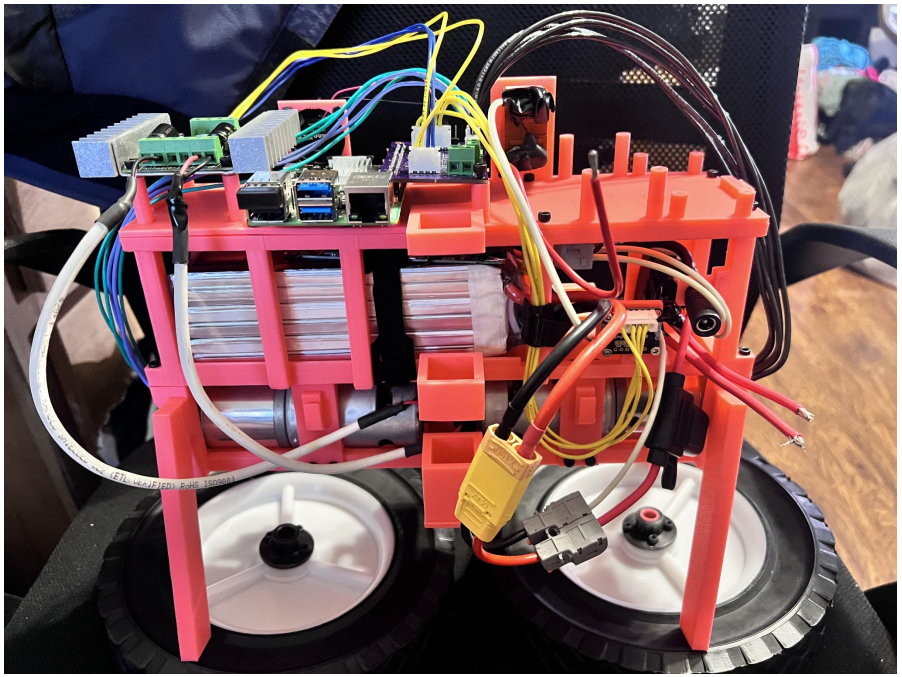
Sean Wall

- Been working to read data from the SQLite database for the app

Chassis & PCB

- Wiring was mostly completed - need to integrate with power boards
- Need to reprint the Robot in a stronger material
- Reordered PCB and Components to allow for full integration





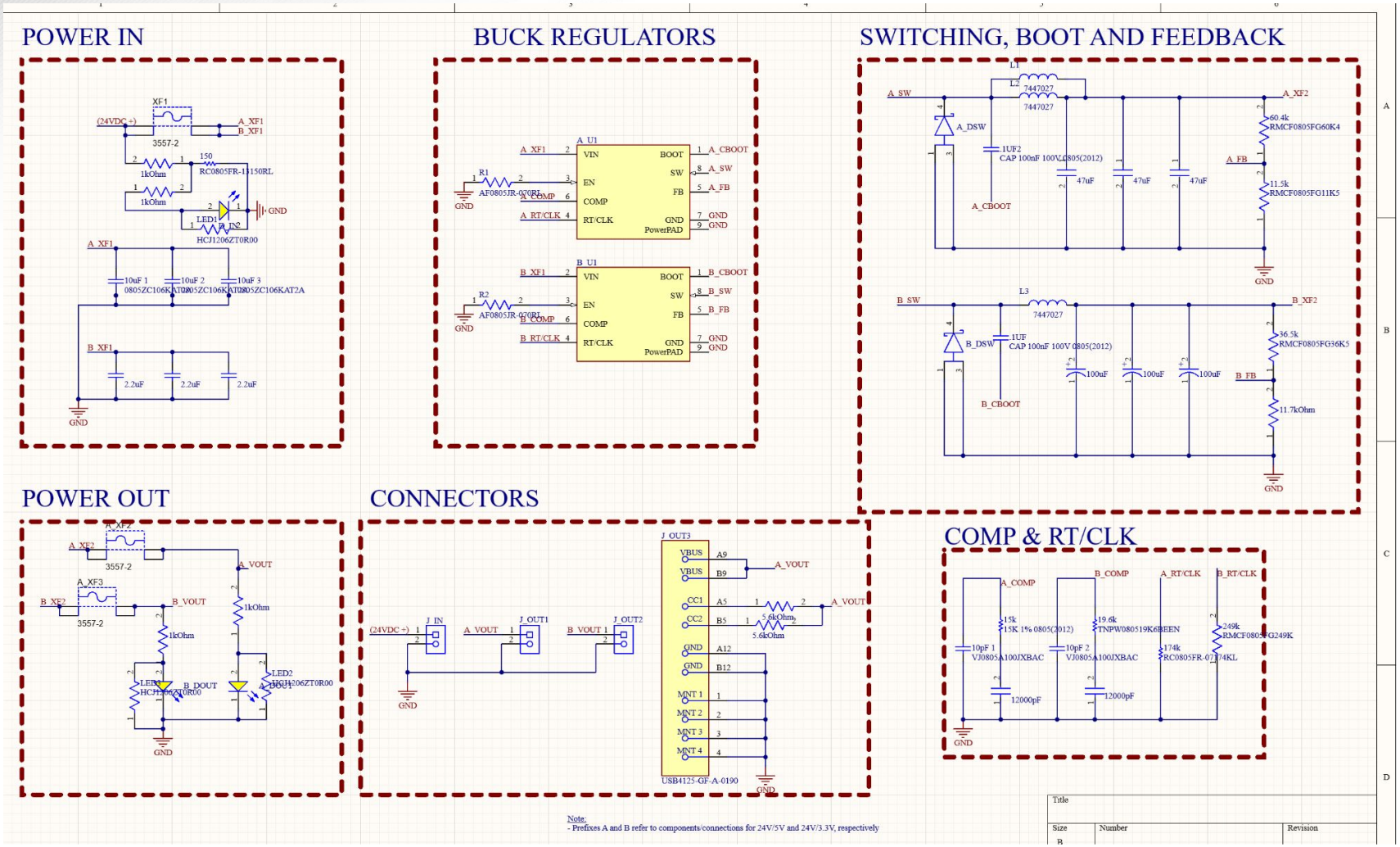
Power System

Stephen Darwin

Accomplishments since last update 10 hrs of effort	Ongoing progress/problems and plans until the next presentation
• Completed simulating/testing for updated buck regulator design • Found appropriate components to correct voltage sag at buck output • Ordered new buck regulator boards and components • Salvaged reusable components from old buck regulator PCBs	• Waiting for boards and components to arrive • Brushing up on soldering with soldering kit • Preparing to integrate with other subsystems • Working on PID control algorithm as backup just in case RL does not work

Power System

Stephen Darwin



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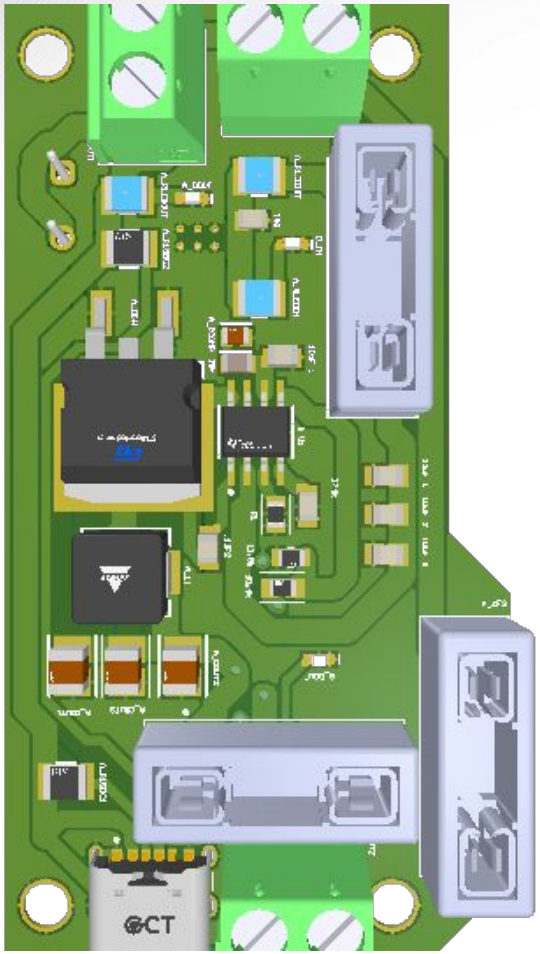
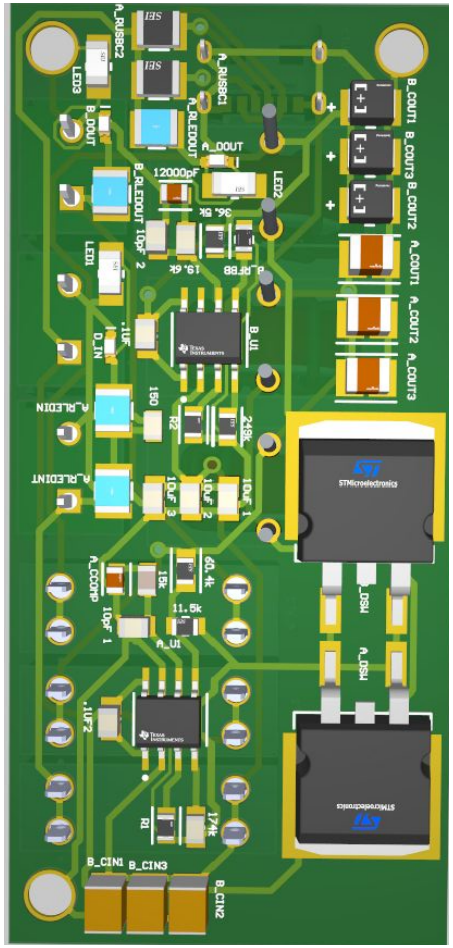
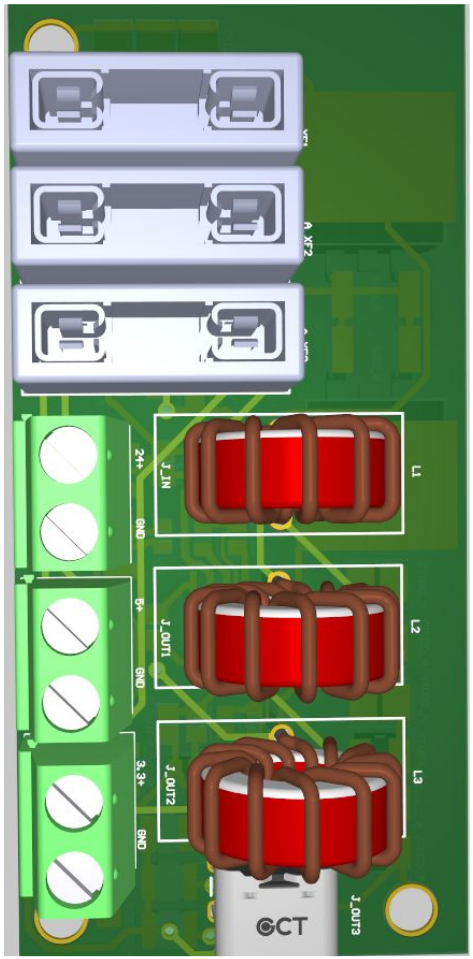
Power System

Stephen Darwin







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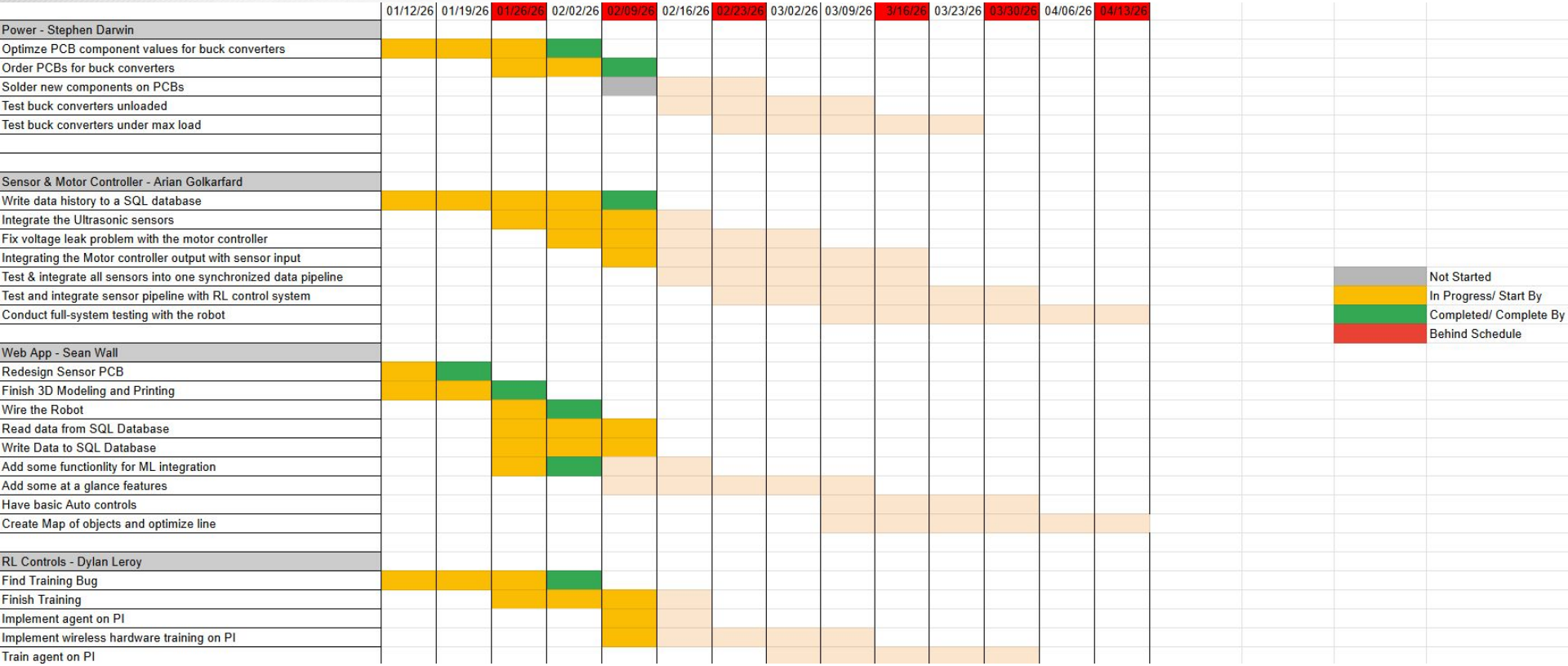
Execution & Plan



Thank You!

Any Questions?